Network Project

Team 3

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1. 물리적 구성도

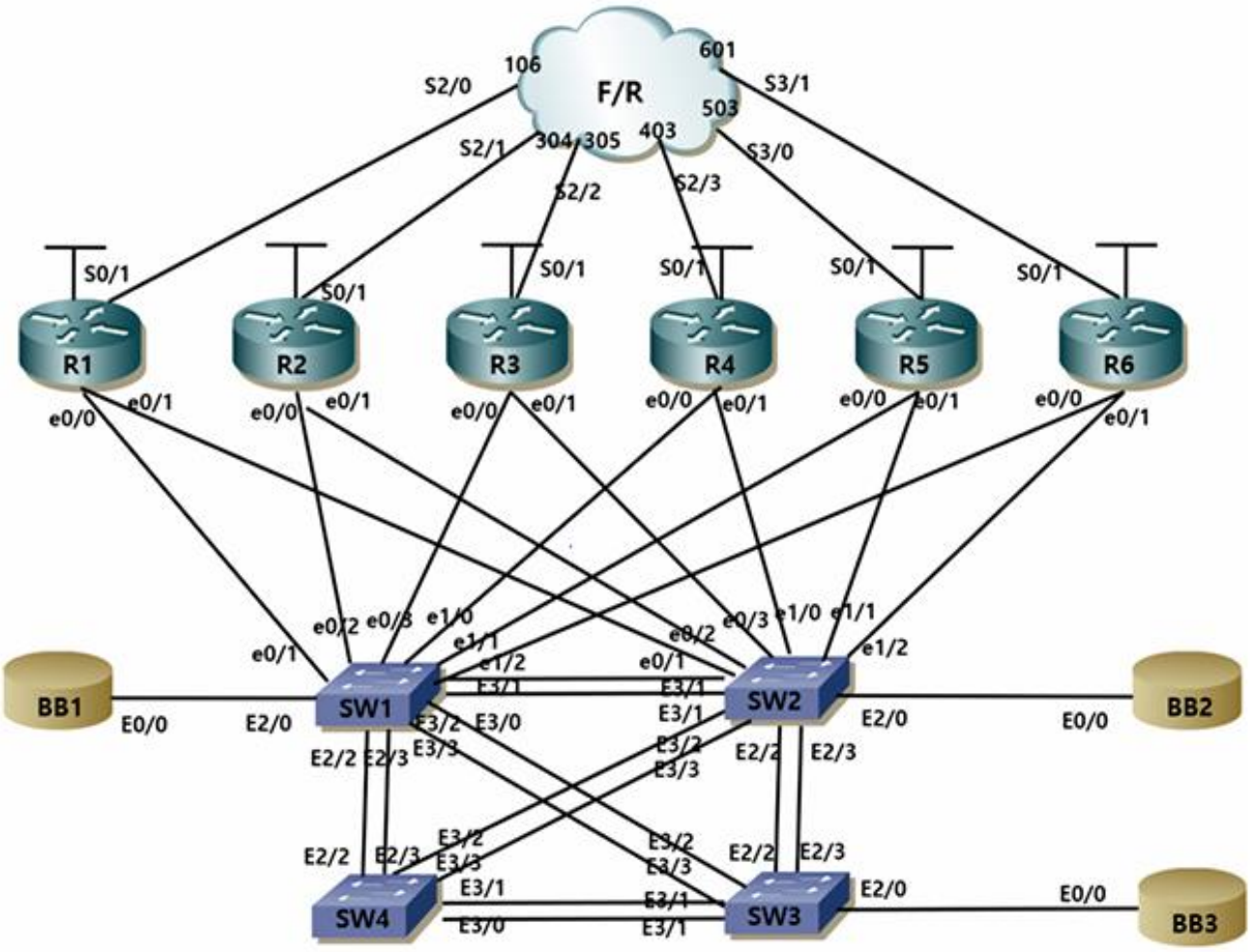
2. IP 구성도

3. IGP 구성도

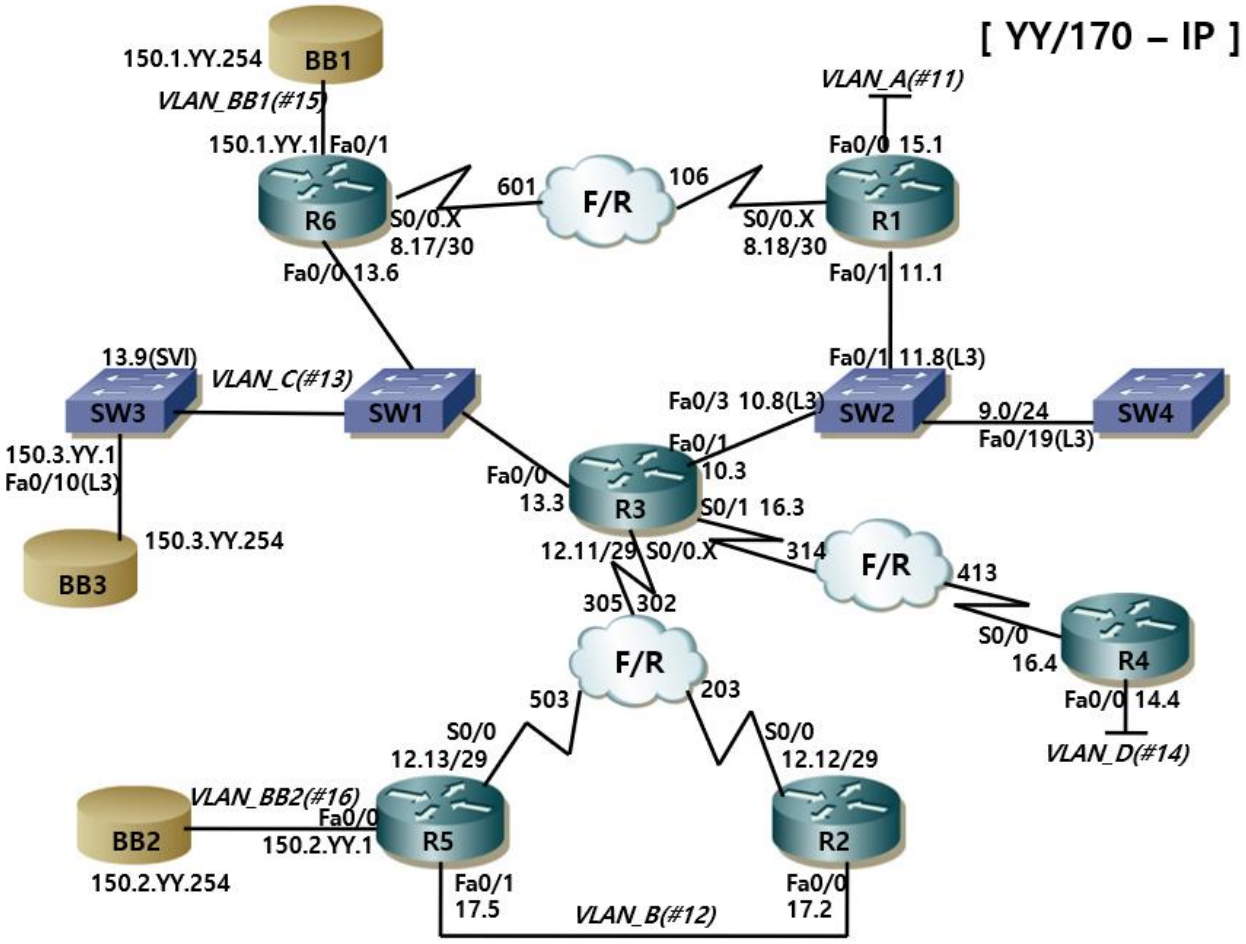
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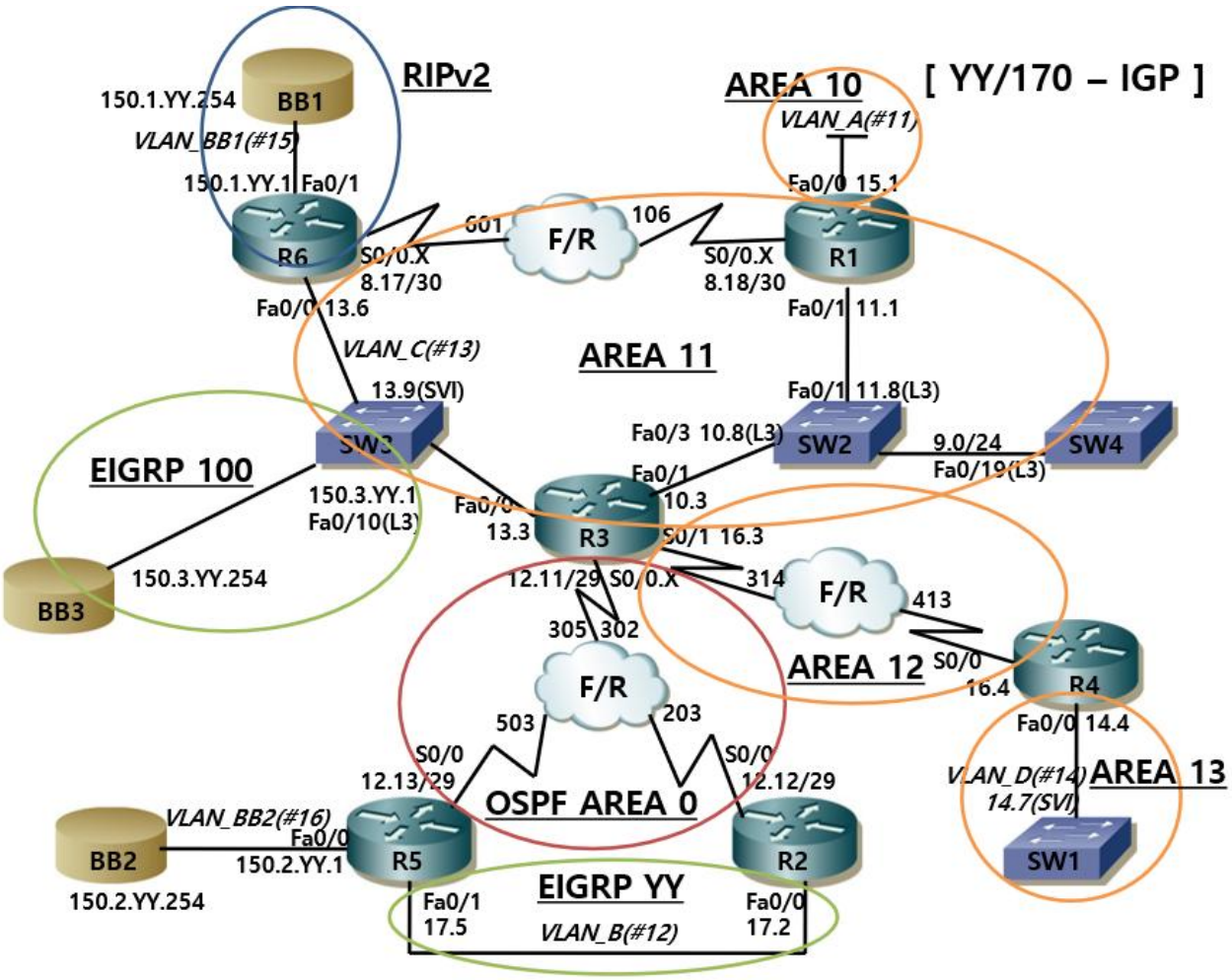
1. 물리적 구성도


F/R
106
601
503
403
304
305
S2/0
S3/1
S2/1
S3/0
S2/2
S2/3
R1
R2
R3
R4
R5
R6
S0/1
e0/0
e0/1
SW1
SW2
SW3
SW4
BB1
BB2
BB3
E0/0
E2/0
E3/0
E3/1
E3/2
E3/3
E2/2
E2/3
e0/1
e0/2
e0/3
e1/0
e1/1
e1/2

2. IP 구성도


[YY/170 – IP]
150.1.YY.254
BB1
VLAN_BB1(#15)
150.1.YY.1 Fa0/1
R6
601
F/R
106
S0/0.X
8.17/30
S0/0.X
8.18/30
R1
VLAN_A(#11)
Fa0/0 15.1
Fa0/0 13.6
Fa0/1 11.1
13.9(SVI)
VLAN_C(#13)
SW3
SW1
Fa0/1 11.8(L3)
Fa0/3 10.8(L3)
SW2
9.0/24
Fa0/19(L3)
SW4
150.3.YY.1
Fa0/10(L3)
Fa0/0
13.3
Fa0/1
10.3
R3
S0/1 16.3
12.11/29 S0/0.X
314
F/R
413
150.3.YY.254
BB3
305 302
F/R
S0/0
16.4
R4
Fa0/0 14.4
VLAN_D(#14)
503
203
S0/0
12.13/29
S0/0
12.12/29
VLAN_BB2(#16)
Fa0/0
BB2
150.2.YY.1
R5
R2
150.2.YY.254
Fa0/1
17.5
VLAN_B(#12)
Fa0/0
17.2

3. IGP 구성도


RIPv2
[YY/170 – IGP]
150.1.YY.254
BB1
VLAN_BB1(#15)
150.1.YY.1 Fa0/1
R6
S0/0.X
8.17/30
Fa0/0 13.6
601
F/R
106
S0/0.X
8.18/30
R1
AREA 10
VLAN_A(#11)
Fa0/0 15.1
Fa0/1 11.1
VLAN_C(#13)
13.9(SVI)
AREA 11
Fa0/1 11.8(L3)
Fa0/3 10.8(L3)
SW2
9.0/24
Fa0/19(L3)
SW4
EIGRP 100
150.3.YY.1
Fa0/10(L3)
Fa0/1
10.3
Fa0/0
13.3
R3
S0/1 16.3
12.11/29 S0/0.X
314
F/R
413
150.3.YY.254
BB3
305 302
F/R
AREA 12
S0/0
16.4
R4
Fa0/0 14.4
503
203
S0/0
12.13/29
S0/0
12.12/29
OSPF AREA 0
VLAN_D(#14)
14.7(SVI)
AREA 13
SW1
VLAN_BB2(#16)
Fa0/0
150.2.YY.1
BB2
150.2.YY.254
R5
R2
EIGRP YY
VLAN_B(#12)
Fa0/1
17.5
Fa0/0
17.2

4. 설정 (170-IMAP)

Ⅰ. Bridging and Switching

1. Configure IP across your frame relay network

[Frame-relay]

[a. R1,R3,R6 을 제외한 나머지 라우터에서는 Sub-interface 를 사용 할 수 없다.]

[b. R1과 R6에서는 Point-to-point sub-interface 를 사용하여 frame-relay 구간 설정.]

[c. R3는 multipoint sub-interface 를 사용하여 frame-relay 구간 설정.]

[d. Sub-interface 를 사용하는 모든 라우터에서는 Sub-interface name 을 Serial0/0.X 로 설정. X=장비번호]

R1

```
int lo0
ip add 23.23.1.1 255.255.255.0

int e0/1
no sh
ip add 23.23.11.1 255.255.255.0

int e0/0
no sh
ip add 23.23.15.1 255.255.255.0

int s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no sh

int s1/0.1 point
ip add 23.23.8.18 255.255.255.252
frame inter 106
```

R2

```
int lo0
ip add 23.23.2.2 255.255.255.0
```

```
int e0/0
no sh
ip add 23.23.17.2 255.255.255.0

int s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no sh
ip add 23.23.12.12 255.255.248
fram map ip 23.23.12.11 203 br
fram map ip 23.23.12.13 203 br
```

R3

```
int lo0
ip add 23.23.3.3 255.255.255.0

int e0/1
no sh
ip add 23.23.10.3 255.255.255.0

int e0/0
no sh
ip add 23.23.13.3 255.255.255.0

int s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no sh

int s1/0.3 multi
ip add 23.23.12.11 255.255.255.248
frame map ip 23.23.12.13 305 broad
frame map ip 23.23.12.12 302 broad

int s1/0.16 p
ip add 23.23.16.3 255.255.255.0
frame int 314
```

R4

```
int lo0
ip add 23.23.4.4 255.255.255.0

int e0/0
no sh
ip add 23.23.14.4 255.255.255.0

int s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no sh

int s1/0.16 p
No sh
ip add 23.23.16.4 255.255.255.0
Fram int 413
```

R5

```
int lo0
ip add 23.23.5.5 255.255.255.0

int e0/0
no sh
ip add 150.2.23.1 255.255.255.0

int e0/1
no sh
ip add 23.23.17.5 255.255.255.0

int s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no sh
ip add 23.23.12.13 255.255.255.248
fram map ip 23.23.12.11 503 br
fram map ip 23.23.12.12 503 br
```

R6

```
int lo0
ip add 23.23.6.6 255.255.255.255

int e0/1
no sh
ip add 150.1.23.1 255.255.255.0

int e0/0
no sh
ip add 23.23.13.6 255.255.255.0

int s1/0
encapsulation frame-relay
no frame-relay inverse-arp
no sh

int s1/0.6 point
ip add 23.23.8.17 255.255.255.252
frame inter 601
```

2. Cat3550 Switch Setup

2.1 Default Configuration

SW1

```
int ran e2/2 - 3
shutdown

int e3/3
shutdown
```

SW2

```
int ran e2/2 – 3
shutdown

int e3/3
shutdown

int e3/2
no sw
```

SW3

```
int ran e3/0 – 1
shutdown

int range e2/2 – 3
shutdown

int e3/3
shutdown
```

SW4

```
int ran e3/0 – 1
shutdown

int ran e2/2 – 3
shutdown
int e3/3
shutdown

int e3/2
no sw
```

2.2 VTP Configuration

[모든 SW에서 VTP 및 DTP broadcast 트래픽이 모든 포트에서 흘러 다니지 않도록 하라.]

SW1 - SW4

```
vtp mode transparent
vtp domain Team3.com
vtp password Team3
vtp ver 2
```

2.3 Trunk Port

[SW1,SW2 사이에 ISL trunk 구성하라.]

SW1

```
int ran e3/0 - 1
switch trunk encap isl
switch mode trunk

int e3/2
switch trunk encap dot
switch mode trunk
```

SW2

```
Int ran e3/0 - 1
switch trunk encap isl
switch mode trunk
```

SW3

```
int e3/2
switch trunk encap dot
switch mode trunk
```

2.4 VLAN Configuration

[다음에 보여지는 것과 같이 VLAN을 assign 하시오. 모든 설정이 완료된 후, SW1,2,3에서 모든 VLAN이 보여야됨]

VLAN name	No
VLAN_A	11
VLAN_B	12
VLAN_C	13
VLAN_D	14
VLAN_BB1	15
VLAN_BB2	16

SW1

```
no Vlan 10-100

vlan 11
name VLAN_A
vlan 12
name VLAN_B
vlan 13
name VLAN_C
vlan 14
name VLAN_D
vlan 15
name VLAN_BB1
vlan 16
name VLAN_BB2



int e0/1
sw mode acc
sw acc vlan 11

int e0/2
sw mod acc
sw acc vlan 15
```

```
int e0/3
sw mod acc
sw acc vlan 13

int e1/0
sw mod acc
sw acc vlan 14

int e1/1
sw mod acc
sw acc vlan 16

int e1/2
sw mod acc
sw acc vlan 13
```

SW2

```
no Vlan 10-100

vlan 11
name VLAN_A
vlan 12
name VLAN_B
vlan 13
name VLAN_C
vlan 14
name VLAN_D
vlan 15
name VLAN_BB1
vlan 16
name VLAN_BB2

int e0/1
no sw
ip add 23.23.11.8 255.255.255.0
```

```
int e0/3
no sw
ip add 23.23.10.8 255.255.255.0

int e3/2
no sw
ip add 23.23.9.8 255.255.255.0

int e1/1
sw mod acc
sw acc vlan 12

int e1/2
sw mod acc
sw acc vlan 15

int e2/0
sw mod acc
sw acc vlan 16
int vlan 16
no sh
ip add 150.2.23.1 255.255.255.0
```

SW3

```
no vlan 10-100

vlan 11
name VLAN_A
vlan 12
name VLAN_B
vlan 13
name VLAN_C
vlan 14
name VLAN_D
vlan 15
name VLAN_BB1
```

```
vlan 16
name VLAN_BB2

int e2/0
no sw
ip add 150.3.23.1 255.255.255.0

int vlan 13
no sh
ip add 23.23.13.9 255.255.255.0
```

SW4

```
int e3/2
no sw
ip add 23.23.9.10 255.255.255.0
```

2.5 Load Balancing (홀수-odd VLAN은 F0/23 , 짝수-even VLAN은 f0/24) SW1,2에 설정

[홀수 VLAN은 F0/23 , 짝수 VLAN은 F0/24 흘러 다니게 하라. 사용하는 VLAN만 흘러다녀야 한다.]

SW1-SW2

```
Int e3/0
Sw tr allowed Vlan 11,13,15
Int e3/1
Sw tr all Vlan 12,14,16
```

2.6 Management Interface

(아래 단계에서는 up down 상태, 2.7 설정 후 up up 상태 됨)

'SW1에서 OSPF 포함될 YY.YY.14.7/24 IP를 설정하며 VLAN_D에 대해서 SW1이 root 가 되도록 설정하라' SW3은 YY.YY.13.9/24를 설정한다.

SW1

```
int vlan 14
no sh
ip add 23.23.14.7 255.255.255.0
spanning-tree vlan 14 priority 0
```

SW3

```
int vlan 13
no sh
ip add 23.23.13.9 255.255.255.0
```

2.7 Switch L2 EtherChannel

[SW1과 SW3 사이에 Access 로 설정하며, 각 Link 사이에 IEEE를 사용한 Etherchannel 을 구성하되, SW3은 IEEE Protocol 을 사용하여 Negotitate를 하여야 하며, SW1은 이에 반응하여 Etherchannel 이 형성되도록 하라. Etherchannel 을 구성함에 있어 Port Number 는 YY를 사용하도록 하라.]

SW1

```
int ran e3/2 - 3
channel-group 23 mode passive
int port-channel 23
sw acc vlan 13
```

SW3

```
int ran e3/2 - 3
channel-group 23 mode active
int port-channel 23
switchport access vlan 13
```

2.8 BPDU Attack

SW1

int e2/0

spanning-tree bpdufilter enable

Sw2

int e2/0

spanning-tree bpdufilter enable

2.9 Spanning-tree Tuning

Sw1

spanning-tree vlan 14 root primary

spanning-tree vlan 14 hello-time 2

spanning-tree vlan 14 max-age 16

spanning-tree vlan 14 forward-time 14

2.10 Errdisable

SW1

errdisable recovery cause udld

errdisable recovery interval 600

Ⅱ. IP IGP Protocols

IGP의 router-id 는 다음과 같이 구성하라. (R1 : 1.1.1.1 , R2 : 2.2.2.2 , R3 : 3.3.3.3 ---)]

1. RIP

1.1 RIPv2 Configuration

[구성도-B를 참조하여 R6의 fa0/1 port 만 포함하고 version 2 정보를 BB1 과 주고 받을 수 있도록 하라]

R6

```
router rip
ver 2
no au
net 150.1.0.0
passive-interface default
no passive-interface e0/1
```

1.2 Adjust Received Update

R6

```
ip access-list standard RIP_IN
permit 199.172.0.0 0.0.11.0
router rip
distribute-list RIP_IN in e0/1
```

1.3 Route Redistribution (OSPF 설정 후 적용 및 확인)

[R6에서 RIP OSPF 상호 재분배 하라. BB1으로는 YY.YY.0.0/16만 보내라]

R6

```
router rip
redistribute ospf 1 metric 3
distribute-list prefix R6=>BB1 out e0/1

int e0/1
ip summary-address rip 23.23.0.0 255.255.0.0
ip prefix-list R6=>BB1 permit 23.23.0.0/16
router ospf 1
redistribute rip subnets
```

2. OSPF (OSPF Router-id X.X.X.X X: router number)

2.1 Basic OSPF Configuration

[a. 아래 조건을 참조하여 OSPF 를 구성하시오]

Area 10	R1 fa0/0
Area 11	R1 fa0/1 S0/0 , R3 fa0/0 fa0/1 , R6 fa0/0 S0/0 , SW2 , SW3 , SW4
Area 12	R3 S0/1 , R4 S0/0
Area 13	R4 fa0/0 , SW1

[b. 위 조건에 만족하여 OSPF 를 구성하되 area 10 과 area 13 에 대한 reachable 를 보장하시오.=virtual link]

[c. OSPF AREA 12 를 구성할 때 Hello interval 10초 , Dead interval 40초가 되도록 조정하고자 한다.

(단 설정을 하는 동안 interval(time)과 직접 관련된 명령어는 사용할 수 없다.)]

[d. OSPF AREA 0 구성 시 ip ospf network 관련 명령어 사용 금지. R2-R3-R5 사이에 구성]

R1

```
router ospf 1
router-id 1.1.1.1
network 23.23.1.1 0.0.0.0 area 10
network 23.23.15.1 0.0.0.0 area 10
network 23.23.11.1 0.0.0.0 area 11
network 23.23.8.18 0.0.0.0 area 11
```

R2

```
router ospf 1
router-id 2.2.2.2
network 23.23.2.2 0.0.0.0 area 0
network 23.23.12.12 0.0.0.0 area 0
int s1/0
ip ospf priority 0
```

R3

```
router ospf 1
router-id 3.3.3.3
net 23.23.3.3 0.0.0.0 a 0
net 23.23.10.3 0.0.0.0 a 11
```

net 23.23.12.11 0.0.0.0 a 0
net 23.23.13.3 0.0.0.0 a 11
net 23.23.16.3 0.0.0.0 a 12
nei 23.23.12.12
nei 23.23.12.13

int s1/0
ip ospf net point-to-point

R4

router ospf 1
router-id 4.4.4.4
net 23.23.4.4 0.0.0.0 a 13
net 23.23.16.4 0.0.0.0 a 12
net 23.23.14.7 0.0.0.0 a 13

int s1/0
ip os net point-to-point

R5

router ospf 1
router-id 5.5.5.5
network 23.23.5.5 0.0.0.0 area 0
network 23.23.12.13 0.0.0.0 area 0

int s1/0
ip ospf priority 0

R6

router ospf 1
router-id 6.6.6.6
network 23.23.6.6 0.0.0.0 area 11
network 23.23.13.6 0.0.0.0 area 11
network 23.23.8.17 0.0.0.0 area 11

SW1

router ospf 1

router-id 7.7.7.7

network 23.23.7.7 0.0.0.0 area 13

network 23.23.14.7 0.0.0.0 area 13

SW2

router ospf 1

router-id 8.8.8.8

network 23.23.8.8 0.0.0.0 area 11

network 23.23.9.8 0.0.0.0 area 11

network 23.23.10.8 0.0.0.0 area 11

network 23.23.11.8 0.0.0.0 area 11

SW3

router ospf 1

router-id 9.9.9.9

network 23.23.9.9 0.0.0.0 area 11

network 23.23.13.9 0.0.0.0 area 11

SW4

router ospf 1

router-id 10.10.10.10

router-id 10.10.10.10 0.0.0.0 a 11

network 23.23.9.10 0.0.0.0 area 11

network 23.23.10.10 0.0.0.0 area 11

2.1-1 virtual-link

R1

```
router ospf 1
area 11 virtual-link 3.3.3.3
```

R3

```
router ospf 1
area 11 virtual-link 1.1.1.1
area 12 virtual-link 4.4.4.4
```

R4

```
router ospf 1
area 12 virtual-link 3.3.3.3
```

2.2 OSPF Stub AREA

R4

```
router ospf 1
area 13 stub no-summary
area 13 default-cost 18
```

---

sw1

```
area 13 stub
```

2.3 OSPF Route Reduction (## 기존 area 11 => area 13 변경 축약)

、OSPF AREA 13 에 포함되는 경로에 대해 summary 를 수행하여 다른 OSPF domain 에서 하나의 경로로 보이게 하라.

R4

```
router ospf 1
area 13 range 23.23.0.0 255.255.240.0
area 13 range 23.23.4.4 255.255.255.255
area 13 range 23.23.7.7 255.255.255.255
```

3. EIGRP

3.1 EIGRP 100 Configuration

[BB3에서 4.1.1.0/24, 128.28.2.0/24, 198.198.1.0/24 만 받아라. 이 정보는 모든 라우터에서 확인이 가능해야 하며 OSPF에서 metric 값이 누적되면서 전달되어야 한다.]

SW3

```
router eigrp 100
eigrp router-id 9.9.9.9
no auto-summary
net 150.3.14.1 0.0.0.0
distribute-list prefix BB3=>SW3 in e2/0
redi ospf 1 met 1 1 1 1 1

Ip prefix-list BB3=>SW3 permit 4.1.1.0/24
Ip prefix-list BB3=>SW3 permit 128.28.2.0/24
Ip prefix-list BB3=>SW3 permit 198.198.1.4/24

router ospf 1
redi eigrp 100 subnets route-map CH-TYPE
route-map CH-TYPE
match ip add prefix-list BB3=>SW3
set metric-type type-1
route-map CH-TYPE 20
```

3.2 EIGRP YY Configuration

R2

```
router eigrp 23
eigrp router-id 2.2.2.2
no auto-summary
net 23.23.17.2 0.0.0.0
```

R5

```
router eigrp 23
eigrp router-id 5.5.5.5
no auto
net 23.23.17.5
```

3.3 EIGRP Contingency

[R2와 R5에서 eigrp 와 ospf 재분배를 수행하되 아래 조건에 만족하게끔 구성을 하라.
R2는 Connected interface 를 제외한 모든 경로에 대해 VLAN_B 를 통하여 확인이 가능하도록 구성하라.]

R2

```
router eigrp 23
redi ospf 1 met 1 1 1 1 1
distance eigrp 90 95

route-map to-ospf deny 10
match tag 100
route-map TO_OSPF permit 20

router os 1
redi ei 23 sub route-map TO_OSPF
```

R5

```
router eigrp 23
redi ospf met 1 1 1 1 1 route-map TO_EIGRP

router ospf 1
redi ei 23 sub

route-map TO_EIGRP
set tag 100
```

Ⅲ. IOS/IP Feature

1. NTP

R3

```
ntp authentication-key 1 md5 cisco
ntp authenticate
ntp trusted-key 1
ntp source lo0
ntp master 2
```

R4

```
ntp server 23.23.3.3 key 1
ntp authentication-key 1 md5 cisco
ntp authenticate
ntp trusted-key 1
ntp source lo0
```

SW2

```
ntp server 23.23.3.3 key 1
ntp authentication-key 1 md5 cisco
ntp authenticate
ntp trusted-key 1
ntp source lo0
```

2. HSRP

R2

```
int e0/0
standby 1 ip 23.23.17.1
standby 1 preempt
```

R5

```
int e0/0
standby 1 ip 23.23.17.1
standby 1 priority 150
```

```
standby 1 preempt
standby 1 track s0/0 100
```

3. SYSLOG

R5

```
logging on
logging trap critical
logging facility local6
logging source-interface lo0
logging host 150.2.23.10
```

Ⅵ. Security

1. Catalyst 3550 Switch Security (맨 마지막 적용)

[R2 와 R5가 연결된 SW에 설정하여 두 라우터의 MAC 만이 허용되도록 한다.
이 정책에 위반하면 traffic 은 Log 를 남기도록 하라. SW가 재시작 되는 경우에도 이에 대한 설정은 남아 있도록 하라
HSRP 의 use-bia , mac-address 는 사용해서는 안된다.]

R2

```
int e0/0
standby 1 mac-address 0000.0000.2222
```

R5

```
int e0/1
standby 1 mac-address 0000.0000.0005
```

SW1

```
int e0/2
sw port-security maximum 2
sw port-security mac-addres 0000.0000.22222
sw port-security
```

SW2

```
int e1/1
sw port-security maximum 2
sw port-security mac-address 0000.0000.0005
sw port-security
```

2. Telnet Feature

SW2

```
access-list 99 permit 150.1.0.0 0.0.255.255
access-list 99 permit 23.23.0.0 0.0.255.255
line vty 0 15
access-class 99 in
```

3. Dynamic Access-list

[텔넷 연결은 최대 10분 이상 지속될 수 없고, telnet 의 idle 이 허용되는 시간은 최대 2 분이다.]

R4

```
username ccie password 0 cisco

ip access-list extened lock
permit tcp 23.23.14.0 0.0.0.255 host 23.23.14.4 eq telnet
dynamic lk timeout 0 permit 23.23.14.0 0.0.0.255 any eq telnet
deny tcp 23.23.14.0 0.0.0.255 any eq telnet
permit ip any any

iine vty 0 4
password cisco
login local
autocommand access-enable host timeout 2
int e0/0
ip access-group LOCK in
```

5. 결과

Ⅰ. 라우팅 테이블 확인

R2 (재분배 전)

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

      23.0.0.0/8 is variably subnetted, 16 subnets, 4 masks
O IA     23.23.1.1/32 [110/85] via 23.23.12.11, 01:03:43, Serial1/0
C        23.23.2.0/24 is directly connected, Loopback0
L        23.23.2.2/32 is directly connected, Loopback0
O        23.23.3.3/32 [110/65] via 23.23.12.11, 01:04:18, Serial1/0
O        23.23.5.5/32 [110/65] via 23.23.12.13, 01:04:08, Serial1/0
O IA     23.23.6.6/32 [110/149] via 23.23.12.11, 00:42:56, Serial1/0
O IA     23.23.8.16/30 [110/148] via 23.23.12.11, 00:42:56, Serial1/0
O IA     23.23.9.0/24 [110/84] via 23.23.12.11, 01:04:18, Serial1/0
O IA     23.23.10.0/24 [110/74] via 23.23.12.11, 01:04:18, Serial1/0
O IA     23.23.11.0/24 [110/84] via 23.23.12.11, 01:04:18, Serial1/0
C        23.23.12.8/29 is directly connected, Serial1/0
L        23.23.12.12/32 is directly connected, Serial1/0
O IA     23.23.13.0/24 [110/74] via 23.23.12.11, 01:04:19, Serial1/0
O IA     23.23.15.0/24 [110/94] via 23.23.12.11, 01:03:44, Serial1/0
C        23.23.17.0/24 is directly connected, Ethernet0/0
L        23.23.17.2/32 is directly connected, Ethernet0/0
      150.1.0.0/24 is subnetted, 1 subnets
O E2     150.1.23.0 [110/20] via 23.23.12.11, 01:04:20, Serial1/0
```

(재분배 후)

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

     23.0.0.0/8 is variably subnetted, 22 subnets, 4 masks
D EX    23.23.1.1/32 [95/1734656] via 23.23.17.5, 00:00:49, Ethernet0/0
C       23.23.2.2/24 is directly connected, Loopback0
D EX    23.23.3.3/32 [95/1734656] via 23.23.17.5, 00:00:49, Ethernet0/0
D EX    23.23.4.4/32 [95/1734656] via 23.23.17.5, 00:00:49, Ethernet0/0
D EX    23.23.5.5/32 [95/1734656] via 23.23.17.5, 00:00:49, Ethernet0/0
D EX    23.23.6.6/32 [95/1734656] via 23.23.17.5, 00:00:49, Ethernet0/0
D EX    23.23.7.7/32 [95/1734656] via 23.23.17.5, 00:00:49, Ethernet0/0
D EX    23.23.8.8/32 [95/1734656] via 23.23.17.5, 00:00:49, Ethernet0/0
D EX    23.23.8.16/30 [95/1734656] via 23.23.17.5, 00:00:49, Ethernet0/0
D EX    23.23.9.0/24 [95/1734656] via 23.23.17.5, 00:00:49, Ethernet0/0
D EX    23.23.9.9/32 [95/1734656] via 23.23.17.5, 00:00:49, Ethernet0/0
D EX    23.23.10.0/24 [95/1734656] via 23.23.17.5, 00:00:49, Ethernet0/0
D EX    23.23.10.10/32 [95/1734656] via 23.23.17.5, 00:00:50, Ethernet0/0
D EX    23.23.11.0/24 [95/1734656] via 23.23.17.5, 00:00:50, Ethernet0/0
C       23.23.12.0/29 is directly connected, Serial1/0
L       23.23.12.2/32 is directly connected, Serial1/0
D EX    23.23.13.0/24 [95/1734656] via 23.23.17.5, 00:00:50, Ethernet0/0
D EX    23.23.14.0/24 [95/1734656] via 23.23.17.5, 00:00:50, Ethernet0/0
D EX    23.23.15.0/24 [95/1734656] via 23.23.17.5, 00:00:50, Ethernet0/0
D EX    23.23.16.0/24 [95/1734656] via 23.23.17.5, 00:00:50, Ethernet0/0
C       23.23.17.0/24 is directly connected, Ethernet0/0
L       23.23.17.2/32 is directly connected, Ethernet0/0
     150.1.0.0/24 is subnetted, 1 subnets
D EX    150.1.23.0 [95/1734656] via 23.23.17.5, 00:00:18, Ethernet0/0
     150.3.0.0/24 is subnetted, 1 subnets
D EX    150.3.23.0 [95/1734656] via 23.23.17.5, 00:00:50, Ethernet0/0
```

R5 (재분배 전)

```
R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

      23.0.0.0/8 is variably subnetted, 16 subnets, 4 masks
O IA     23.23.1.1/32 [110/85] via 23.23.12.11, 01:05:35, Serial1/0
O        23.23.2.2/32 [110/65] via 23.23.12.12, 01:06:00, Serial1/0
O        23.23.3.3/32 [110/65] via 23.23.12.11, 01:06:10, Serial1/0
C        23.23.5.0/24 is directly connected, Loopback0
L        23.23.5.5/32 is directly connected, Loopback0
O IA     23.23.6.6/32 [110/149] via 23.23.12.11, 00:44:48, Serial1/0
O IA     23.23.8.16/30 [110/148] via 23.23.12.11, 00:44:48, Serial1/0
O IA     23.23.9.0/24 [110/84] via 23.23.12.11, 01:06:10, Serial1/0
O IA     23.23.10.0/24 [110/74] via 23.23.12.11, 01:06:10, Serial1/0
O IA     23.23.11.0/24 [110/84] via 23.23.12.11, 01:06:10, Serial1/0
C        23.23.12.8/29 is directly connected, Serial1/0
L        23.23.12.13/32 is directly connected, Serial1/0
O IA     23.23.13.0/24 [110/74] via 23.23.12.11, 01:06:12, Serial1/0
O IA     23.23.15.0/24 [110/94] via 23.23.12.11, 01:05:37, Serial1/0
C        23.23.17.0/24 is directly connected, Ethernet0/1
L        23.23.17.5/32 is directly connected, Ethernet0/1
      150.1.0.0/24 is subnetted, 1 subnets
O E2     150.1.23.0 [110/20] via 23.23.12.11, 01:06:13, Serial1/0
      150.2.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        150.2.23.0/24 is directly connected, Ethernet0/0
L        150.2.23.1/32 is directly connected, Ethernet0/0
```

(재분배 후)

```
R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

     23.0.0.0/8 is variably subnetted, 22 subnets, 4 masks
O IA    23.23.1.1/32 [110/85] via 23.23.12.3, 00:10:46, Serial1/0
O       23.23.2.2/32 [110/65] via 23.23.12.2, 00:10:46, Serial1/0
O       23.23.3.3/32 [110/65] via 23.23.12.3, 00:10:46, Serial1/0
O IA    23.23.4.4/32 [110/129] via 23.23.12.3, 00:10:46, Serial1/0
C       23.23.5.5/24 is directly connected, Loopback0
O IA    23.23.6.6/32 [110/75] via 23.23.12.3, 00:10:46, Serial1/0
O IA    23.23.7.7/32 [110/139] via 23.23.12.3, 00:10:46, Serial1/0
O IA    23.23.8.8/32 [110/75] via 23.23.12.3, 00:10:46, Serial1/0
O IA    23.23.8.16/30 [110/138] via 23.23.12.3, 00:10:46, Serial1/0
O IA    23.23.9.0/24 [110/84] via 23.23.12.3, 00:10:46, Serial1/0
O IA    23.23.9.9/32 [110/75] via 23.23.12.3, 00:10:46, Serial1/0
O IA    23.23.10.0/24 [110/74] via 23.23.12.3, 00:10:46, Serial1/0
O IA    23.23.10.10/32 [110/85] via 23.23.12.3, 00:10:46, Serial1/0
O IA    23.23.11.0/24 [110/84] via 23.23.12.3, 00:10:46, Serial1/0
C       23.23.12.0/29 is directly connected, Serial1/0
L       23.23.12.5/32 is directly connected, Serial1/0
O IA    23.23.13.0/24 [110/74] via 23.23.12.3, 00:10:46, Serial1/0
O IA    23.23.14.0/24 [110/138] via 23.23.12.3, 00:10:46, Serial1/0
O IA    23.23.15.0/24 [110/94] via 23.23.12.3, 00:10:46, Serial1/0
O IA    23.23.16.0/24 [110/128] via 23.23.12.3, 00:10:46, Serial1/0
C       23.23.17.0/24 is directly connected, Ethernet0/1
L       23.23.17.5/32 is directly connected, Ethernet0/1
     150.1.0.0/24 is subnetted, 1 subnets
O E2    150.1.23.0 [110/20] via 23.23.12.3, 00:01:10, Serial1/0
     150.2.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       150.2.23.0/24 is directly connected, Ethernet0/0
L       150.2.23.1/32 is directly connected, Ethernet0/0
     150.3.0.0/24 is subnetted, 1 subnets
O E2    150.3.23.0 [110/20] via 23.23.12.3, 00:02:37, Serial1/0
```

R6 (재분배 전)

```
R6#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

      23.0.0.0/8 is variably subnetted, 16 subnets, 4 masks
O IA     23.23.1.1/32 [110/65] via 23.23.8.18, 00:45:32, Serial1/0.6
O IA     23.23.2.2/32 [110/149] via 23.23.8.18, 00:45:32, Serial1/0.6
O IA     23.23.3.3/32 [110/85] via 23.23.8.18, 00:45:32, Serial1/0.6
O IA     23.23.5.5/32 [110/149] via 23.23.8.18, 00:45:32, Serial1/0.6
C        23.23.6.0/24 is directly connected, Loopback0
L        23.23.6.6/32 is directly connected, Loopback0
C        23.23.8.16/30 is directly connected, Serial1/0.6
L        23.23.8.17/32 is directly connected, Serial1/0.6
O        23.23.9.0/24 [110/84] via 23.23.8.18, 00:45:32, Serial1/0.6
O        23.23.10.0/24 [110/84] via 23.23.8.18, 00:45:32, Serial1/0.6
O        23.23.11.0/24 [110/74] via 23.23.8.18, 00:45:32, Serial1/0.6
O IA     23.23.12.8/29 [110/148] via 23.23.8.18, 00:45:32, Serial1/0.6
C        23.23.13.0/24 is directly connected, Ethernet0/0
L        23.23.13.6/32 is directly connected, Ethernet0/0
O IA     23.23.15.0/24 [110/74] via 23.23.8.18, 00:45:33, Serial1/0.6
O E2     23.23.17.0/24 [110/20] via 23.23.8.18, 00:45:24, Serial1/0.6
      150.1.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        150.1.23.0/24 is directly connected, Ethernet0/1
L        150.1.23.1/32 is directly connected, Ethernet0/1
```

(재분배 후)

```
R6#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, + - replicated route

Gateway of last resort is not set

     23.0.0.0/8 is variably subnetted, 22 subnets, 4 masks
O       23.23.1.1/32 [110/31] via 23.23.13.3, 00:12:38, Ethernet0/0
O IA    23.23.2.2/32 [110/75] via 23.23.13.3, 00:11:08, Ethernet0/0
O IA    23.23.3.3/32 [110/11] via 23.23.13.3, 00:12:38, Ethernet0/0
O IA    23.23.4.4/32 [110/75] via 23.23.13.3, 00:12:38, Ethernet0/0
O IA    23.23.5.5/32 [110/75] via 23.23.13.3, 00:11:08, Ethernet0/0
C       23.23.6.6/24 is directly connected, Loopback0
O IA    23.23.7.7/32 [110/85] via 23.23.13.3, 00:12:38, Ethernet0/0
O       23.23.8.8/32 [110/21] via 23.23.13.3, 00:12:38, Ethernet0/0
C       23.23.8.16/30 is directly connected, Serial1/0.6
L       23.23.8.17/32 is directly connected, Serial1/0.6
O       23.23.9.0/24 [110/30] via 23.23.13.3, 00:12:38, Ethernet0/0
O       23.23.9.9/32 [110/11] via 23.23.13.9, 00:12:38, Ethernet0/0
O       23.23.10.0/24 [110/20] via 23.23.13.3, 00:12:39, Ethernet0/0
O       23.23.10.10/32 [110/31] via 23.23.13.3, 00:12:39, Ethernet0/0
O       23.23.11.0/24 [110/30] via 23.23.13.3, 00:12:39, Ethernet0/0
O IA    23.23.12.0/29 [110/74] via 23.23.13.3, 00:12:39, Ethernet0/0
C       23.23.13.0/24 is directly connected, Ethernet0/0
L       23.23.13.6/32 is directly connected, Ethernet0/0
O IA    23.23.14.0/24 [110/84] via 23.23.13.3, 00:12:39, Ethernet0/0
O IA    23.23.15.0/24 [110/40] via 23.23.13.3, 00:12:39, Ethernet0/0
O IA    23.23.16.0/24 [110/74] via 23.23.13.3, 00:12:39, Ethernet0/0
O E2    23.23.17.0/24 [110/20] via 23.23.13.3, 00:11:04, Ethernet0/0
     150.1.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       150.1.23.0/24 is directly connected, Ethernet0/1
L       150.1.23.1/32 is directly connected, Ethernet0/1
     150.3.0.0/24 is subnetted, 1 subnets
O E2    150.3.23.0 [110/20] via 23.23.13.9, 00:03:05, Ethernet0/0
```

SW3 (재분배 전)

```
SW3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

      23.0.0.0/8 is variably subnetted, 16 subnets, 4 masks
O IA     23.23.1.1/32 [110/11] via 23.23.11.1, 01:21:36, Ethernet0/1
O IA     23.23.2.2/32 [110/75] via 23.23.10.3, 01:21:51, Ethernet0/3
O IA     23.23.3.3/32 [110/11] via 23.23.10.3, 01:22:33, Ethernet0/3
O IA     23.23.5.5/32 [110/75] via 23.23.10.3, 01:21:51, Ethernet0/3
O        23.23.6.6/32 [110/75] via 23.23.11.1, 01:00:28, Ethernet0/1
O        23.23.8.16/30 [110/74] via 23.23.11.1, 01:22:33, Ethernet0/1
C        23.23.9.0/24 is directly connected, Ethernet3/2
L        23.23.9.8/32 is directly connected, Ethernet3/2
C        23.23.10.0/24 is directly connected, Ethernet0/3
L        23.23.10.8/32 is directly connected, Ethernet0/3
C        23.23.11.0/24 is directly connected, Ethernet0/1
L        23.23.11.8/32 is directly connected, Ethernet0/1
O IA     23.23.12.8/29 [110/74] via 23.23.10.3, 01:22:33, Ethernet0/3
O        23.23.13.0/24 [110/20] via 23.23.10.3, 01:22:33, Ethernet0/3
O IA     23.23.15.0/24 [110/20] via 23.23.11.1, 01:21:36, Ethernet0/1
O E2     23.23.17.0/24 [110/20] via 23.23.10.3, 01:15:57, Ethernet0/3
      150.1.0.0/24 is subnetted, 1 subnets
O E2     150.1.23.0 [110/20] via 23.23.11.1, 01:00:28, Ethernet0/1
      150.3.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        150.3.14.0/24 is directly connected, Ethernet2/0
L        150.3.14.1/32 is directly connected, Ethernet2/0
```

(재분배 후)

```
SW3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

     23.0.0.0/8 is variably subnetted, 21 subnets, 4 masks
O       23.23.1.1/32 [110/22] via 23.23.13.3, 00:13:40, Vlan13
O IA    23.23.2.2/32 [110/66] via 23.23.13.3, 00:11:35, Vlan13
O IA    23.23.3.3/32 [110/2] via 23.23.13.3, 00:13:40, Vlan13
O IA    23.23.4.4/32 [110/66] via 23.23.13.3, 00:13:30, Vlan13
O IA    23.23.5.5/32 [110/66] via 23.23.13.3, 00:11:35, Vlan13
O       23.23.6.6/32 [110/2] via 23.23.13.6, 00:13:05, Vlan13
O IA    23.23.7.7/32 [110/76] via 23.23.13.3, 00:13:30, Vlan13
O       23.23.8.8/32 [110/12] via 23.23.13.3, 00:13:40, Vlan13
O       23.23.8.16/30 [110/65] via 23.23.13.6, 00:13:05, Vlan13
O       23.23.9.0/24 [110/21] via 23.23.13.3, 00:13:40, Vlan13
C       23.23.9.9/24 is directly connected, Loopback0
O       23.23.10.0/24 [110/11] via 23.23.13.3, 00:13:40, Vlan13
O       23.23.10.10/32 [110/22] via 23.23.13.3, 00:13:40, Vlan13
O       23.23.11.0/24 [110/21] via 23.23.13.3, 00:13:40, Vlan13
O IA    23.23.12.0/29 [110/65] via 23.23.13.3, 00:13:40, Vlan13
C       23.23.13.0/24 is directly connected, Vlan13
L       23.23.13.9/32 is directly connected, Vlan13
O IA    23.23.14.0/24 [110/75] via 23.23.13.3, 00:13:30, Vlan13
O IA    23.23.15.0/24 [110/31] via 23.23.13.3, 00:13:40, Vlan13
O IA    23.23.16.0/24 [110/65] via 23.23.13.3, 00:13:40, Vlan13
O E2    23.23.17.0/24 [110/20] via 23.23.13.3, 00:11:30, Vlan13
     150.1.0.0/24 is subnetted, 1 subnets
O E2    150.1.23.0 [110/20] via 23.23.13.6, 00:02:04, Vlan13
     150.3.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       150.3.23.0/24 is directly connected, Ethernet2/0
L       150.3.23.1/32 is directly connected, Ethernet2/0
```

Ⅱ. VTP 정보 확인

SW1

```
VTP Version capable             : 1 to 3
VTP version running             : 2
VTP Domain Name                 : Team3.com
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : aabb.cc00.0700
Configuration last modified by 0.0.0.0 at 2-27-26 03:12:05

Feature VLAN:
--------------
VTP Operating Mode                : Transparent
Maximum VLANs supported locally   : 1005
Number of existing VLANs          : 17
Configuration Revision            : 0
MD5 digest                        : 0x01 0xCA 0x78 0xC6 0x0B 0x1F 0x9F 0x23
                                    0x6D 0x56 0x1E 0x13 0x2E 0x71 0xEC 0x04
```

SW2

```
VTP Version capable             : 1 to 3
VTP version running             : 2
VTP Domain Name                 : Team3.com
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : aabb.cc00.0800
Configuration last modified by 0.0.0.0 at 2-27-26 03:12:05

Feature VLAN:
--------------
VTP Operating Mode                : Transparent
Maximum VLANs supported locally   : 1005
Number of existing VLANs          : 12
Configuration Revision            : 0
MD5 digest                        : 0xCE 0xDF 0x0F 0xBD 0x59 0x5E 0xF0 0xE5
                                    0x92 0xB0 0x3A 0xFF 0x5F 0x4F 0x02 0x9F
```

SW3

```
VTP Version capable             : 1 to 3
VTP version running             : 2
VTP Domain Name                 : Team3.com
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : aabb.cc00.0900
Configuration last modified by 0.0.0.0 at 2-27-26 03:12:05

Feature VLAN:
--------------
VTP Operating Mode                : Transparent
Maximum VLANs supported locally   : 1005
Number of existing VLANs          : 12
Configuration Revision            : 0
MD5 digest                        : 0xCE 0xDF 0x0F 0xBD 0x59 0x5E 0xF0 0xE5
                                    0x92 0xB0 0x3A 0xFF 0x5F 0x4F 0x02 0x9F
```

SW4

```
VTP Version capable             : 1 to 3
VTP version running             : 2
VTP Domain Name                 : Team3.com
VTP Pruning Mode                : Disabled
VTP Traps Generation            : Disabled
Device ID                       : aabb.cc00.0a00
Configuration last modified by 0.0.0.0 at 2-26-26 04:43:53

Feature VLAN:
--------------
VTP Operating Mode                : Transparent
Maximum VLANs supported locally   : 1005
Number of existing VLANs          : 11
Configuration Revision            : 0
MD5 digest                        : 0x0C 0x13 0x56 0xF0 0xBC 0x57 0xEA 0xF6
                                    0x83 0xB4 0x3E 0x8D 0xCA 0xA7 0xAE 0xC2
```

Ⅲ. VLAN 정보 확인

SW1

```
SW1#sh vl br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et0/0, Et0/3, Et1/3, Et2/1
                                                Et2/2, Et2/3, Et3/3
11   VLAN_A                           active    Et0/1
12   VLAN_B                           active    Et0/2
13   VLAN_C                           active    Et1/2, Et3/2
14   VLAN_D                           active    Et1/0
15   VLAN_BB1                         active    Et2/0
16   VLAN_BB2                         active    Et1/1
20   VLAN0020                         active
30   VLAN0030                         active
40   VLAN0040                         active
78   VLAN0078                         active
123  VLAN0123                         active
1002 fddi-default                     act/unsup
1003 trcrf-default                    act/unsup
1004 fddinet-default                  act/unsup
1005 trbrf-default                    act/unsup
```

SW2

```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Et0/0, Et0/2, Et1/0, Et1/3
                                                Et2/1, Et2/2, Et2/3, Et3/3
11   VLAN_A                           active
12   VLAN_B                           active    Et1/1
13   VLAN_C                           active
14   VLAN_D                           active
15   VLAN_BB1                         active    Et1/2
16   VLAN_BB2                         active    Et2/0
1002 fddi-default                     act/unsup
1003 trcrf-default                    act/unsup
1004 fddinet-default                  act/unsup
1005 trbrf-default                    act/unsup
```

SW3

```
SW3#sh vl br

VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Et0/0, Et0/1, Et0/2, Et0/3
                                                 Et1/0, Et1/1, Et1/2, Et1/3
                                                 Et2/1, Et2/2, Et2/3, Et3/0
                                                 Et3/1, Et3/3
11   VLAN_A                            active
12   VLAN_B                            active
13   VLAN_C                            active
14   VLAN_D                            active
15   VLAN_BB1                          active
16   VLAN_BB2                          active
1002 fddi-default                      act/unsup
1003 trcrf-default                     act/unsup
1004 fddinet-default                   act/unsup
1005 trbrf-default                     act/unsup
```

감사합니다.

Team 3

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